

Logging and the ELK Stack

Can you imagine debugging any issue without seeing a log on the production system? The simple answer is no, as it would be difficult to go back in time. Therefore, we need logging. Logs also give us warning signals about the system if they are designed and coded that way. Logging and log analysis is an important step for troubleshooting any issue, and also for throughput, capacity, and monitoring the health of the system. Therefore, having a very good logging platform and strategy will enable effective debugging. Logging is one of the most important key components of software development in the initial stages.

Microservices are generally deployed using image containers such as Docker that provide the log with commands that help you read the logs of services that are deployed inside containers. Docker and Docker Compose provide commands to stream the log output of running services within the container and in all containers, respectively:

Please refer to the following logs command of Docker and Docker Compose:

Docker logs command:

Usage: `docker logs [OPTIONS] <CONTAINER NAME>`

Fetch the logs of a container:

`-f, --follow` Follow log output

`--help` Print usage

`--since=""` Show logs since timestamp

`-t, --timestamps` Show timestamps

`--tail="all"` Number of lines to show from the end of the logs

Docker Compose logs command:

Usage: `docker-compose logs [options] [SERVICE...]`

Options:

`--no-color` Produce monochrome output

`-f, --follow` Follow log output

`-t, --timestamps` Show timestamps

`--tail` Number of lines to show from the end of the logs for each container

[SERVICES...] Service representing the container - you can give multiple

These commands help you to explore the logs of microservices and other processes running inside the containers. As you can see, using the preceding commands would be a challenging task when you have a higher number of services. For example, if you have tens or hundreds of microservices, it would be very difficult to track each microservice log. Similarly, you can imagine, even without containers, how difficult it would be to monitor logs individually. Therefore, you can assume the difficulty of exploring and correlating the logs of tens to hundreds of containers. It is time-consuming and adds very little value.

Therefore, a log aggregator and visualizing tools such as the ELK Stack come to our rescue. It will be used for centralizing logging. We'll explore this in the next section.

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